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An Explainable and Privacy-Preserving Federated Learning Model for Threat Detection in Cyber-Physical-Social Systems

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***Abstract*—Cyber-physical-social systems (CPSS), the technological backbone of infrastructures such as energy, aerospace, transportation, and human-centric applications, are increasingly prone to cyberattacks due to their interconnected nature. Existing threat detection models focus on performance metrics, while effective, often overlook the reliability and privacy of data in CPSS and lack explainability in detecting anomalies. This article introduces an innovative federated learning (FL) model in threat detection for the resilience assessment and enhancement of CPSS. It is not only explainable and privacy-preserving (PP) but also ensures the reliable operation of systems by addressing their inherent vulnerabilities. It addresses current shortcomings, emphasizing the often-neglected accuracy and reliability in literature. In this regard, we reinforce privacy by utilizing differential privacy (DP) and partially homomorphic encryption (PHE), especially the Paillier homomorphic cryptosystem, for secure model update aggregation in the FL environment. The system's resilience against malicious attempts is heightened by ensuring a reliable aggregation process. To enhance model transparency and facilitate effective responses to threats, we integrate SHapley Additive exPlanations (SHAP) for detailed predictive contribution breakdowns and discern relationships between CPSS' sensors, cyber components, and social interactions for aberration detection. This improves system reliability by clarifying the contribution of each sensor, device, or social actor, and identifying potential points of failure. The efficacy of our model was rigorously evaluated on three representative CPSS datasets: Secure Water Treatment (SWaT), Gas Pipeline, and UNSW-NB15. It achieved anomaly detection accuracies of 97.46%, 98.85%, and 93.60%, respectively, outperforming several baseline methods. Our model maintains strong privacy guarantees through cryptographic techniques and provides meaningful explainability during anomaly detection, fostering greater trust and reliability in critical CPSS operations.**

***Index Terms*—Cyber-physical-social systems (CPSS), explainable AI, federated learning (FL), privacy preserving, reliability, threats.**

I. Introduction

CYBER-PHYSICAL-SOCIAL system (CPSS) underpin critical infrastructure sectors such as energy, aerospace, transportation, and human-centric applications. These complex and interconnected systems, responsible for managing intricate processes across cyber, physical, and social domains, are pivotal to modern civilization's seamless functioning. However, their importance necessitates robust security measures and reliable operations. The rise of the Internet of Things (IoT) has amplified the relevance of CPSS, but it has also heightened their vulnerability to cyber threats [1], [2]. The interconnectivity, the essence of IoT, grants CPSS unmatched functionality and precision. Yet, this very feature enlarges the attack surface for cybercriminals, potentially revealing numerous attack vectors [3]. Furthermore, any disruption in their operation can lead to compromised reliability, with downstream effects on the systems they govern. The 2022 Belarusian Railway cyberattack [4], [5] underscored this vulnerability, showing the devastating societal impact of CPSS disruptions. Given the evolving threat landscape, threat and anomaly detection are central to resilience assessment and ensuring the reliability and enhancement of CPSS. Advanced machine learning (ML) techniques are being championed as potent countermeasures. These techniques discern and counteract system anomalies by recognizing standard operational behaviors and flagging significant deviations [6]. ML models, renowned for their pattern recognition capabilities, show promise in identifying attack behaviors in CPSS environments [7], [8]. Ensuring reliable and consistent system operations, especially amidst cyber threats, is crucial. While much literature extolls ML-based models, assessing them using metrics such as precision, recall, and F-measure [9], there is a research gap concerning their accuracy in CPSS threat detection. Moreover, critical aspects such as explainability, reliability, and

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privacy in these models are often neglected, leading to potential closed box issues, privacy breaches, and inconsistent system behavior [10], [11]. Addressing these challenges is paramount for successfully implementing ML in the resilience assessment and enhancement of CPSS. These challenges can be grouped into three main areas.

1) *Accuracy and Reliability:* Despite extensive research on ML for CPSS attack detection, most studies focus on metrics such as precision, recall, and F1 score, often sidelining accuracy [6]. In many instances, models with high precision or recall might compromise accuracy, diminishing their real-world efficacy and potentially leading to unreliable system operations.
2) Model Explainability: Complex ML models, especially deep neural networks, suffer from a lack of transparency, often called closed boxes. For stakeholders, comprehending these models' decision-making is vital for trust, reliable interventions, and effective threat response [9], [12].
3) Data Privacy: CPSS' distributed architecture and the sensitive data they process demand stringent privacy measures. Many ML models fall short in this aspect, potentially revealing critical data to unauthorized users, thereby amplifying security risks [10].

Reliability, security, and privacy are foundational for CPSS operations. A breach in these areas could lead to system failures or severe outcomes. To bolster the CPSS ecosystem's reliability and security, our work focuses on enhancing anomaly detection, emphasizing explainability and data privacy in ML models. With evolving technological threats, we have prioritized creating robust CPSS threat detection models that prioritize both transparency and privacy. Reliability in CPSS necessitates models that can consistently and accurately identify threats. Addressing ML's explainability challenge, especially in CPSS, led us to SHapley Additive exPlanations (SHAP) [13], a premier tool in explainable artificial intelligence (XAI). SHAP, rooted in cooperative game theory, quantifies feature significance in relation to predictions, clarifying their roles in model decisions. For CPSS, SHAP elucidates decision-making, fosters trust, offers actionable insights, and reinforces our pursuit of transparent and reliable ML-based CPSS anomaly detection. In the CPSS landscape, privacy and reliability are paramount. Federated learning (FL) [14], with its decentralized nature, is a notable solution. Through FL, local devices process learning, and only model updates are shared with a central server, minimizing data exposure risks. Indeed, as mentioned in [15], FL approaches offer significant advantages in learning strategies and optimization techniques in a federated setting. Our objective is to utilize the potential of FL to create a reliable anomaly detection model while safeguarding CPSS data privacy. Recent advancements have extended FL's applicability to complex domains such as intelligent transportation and medical systems [16], [17].

However, even with FL's inherent privacy advantages, vulnerabilities such as inversion and inference attacks [18] remain. This article introduces an explainable and privacy-preserving FL (PPFL) model designed for CPSS threat detection. We integrate FL with SHAP for transparency and enhance privacy via cryptographic techniques such as differential privacy (DP) and partially homomorphic encryption (PHE) [19]. Our model analyzes CPSS sensor patterns, cyber data, and social interactions, fortifying data privacy using FL, encryption, and DP. Furthermore, Paillier homomorphic cryptosystems (PHC) [19] ensure reliable and secure gradient aggregation. In essence, our contributions underscore the balance between privacy, reliability, and efficiency in CPSS.

1) We design and implement a novel explainable and PPFL model for CPSS threat detection. This model leverages decentralized data, ensuring privacy and enhancing detection reliability by ensuring potential threats do not exploit centralized vulnerabilities.
2) Incorporation of DP into FL, offering mathematical privacy guarantees. We also utilize PHE to protect model updates during FL by encrypting them. This dual strategy safeguards data even during breaches, as genuine model parameters stay concealed. We balance data privacy, reliability, and accurate detection by adeptly adjusting privacy parameters and enhancing FL's security measures.
3) Augmentation of our model's explainability via SHAP integration. SHAP elucidates the influence of input features on predictions, granting a clear and intuitive understanding of the decision-making process. This fosters system trust, ensures more reliable threat assessment, and empowers operators to address detected threats effectively.
4) A comprehensive analysis comparing the effects of DP and PHE on performance metrics, shedding light on their impact on the model's efficacy and guiding the choice of privacy-enhancing techniques based on CPSS needs.

This article is structured as follows. Section II discusses related literature. Section III defines the problem and details the threat model. Section IV describes our proposed privacy-preserving (PP) model, and Section V details an in-depth security analysis. Section VI assesses our model against prior works. Section VII concludes the article.

II. Related Works

The burgeoning field of threat detection in CPSS and industrial CPS (ICPS) has garnered significant attention, catalyzing scholarly pursuits to address prevalent security challenges through innovative technological solutions [20]. Among these, ML, DL, and FL have emerged as pivotal methodologies underpinned by the need for enhanced model explainability. In an exemplar of DL innovation, [21] introduced an ensemble model combining long short-term memory with auto-encoder architectures, adept at learning standard data patterns while effectively reducing dimensionality. This model exhibited exceptional accuracies of 99.7% and 99.3% on the Secure Water Treatment (SWaT) and Gas Pipeline datasets, respectively. Further addressing the imbalance in ICPS data, a convolutional neural network, augmented with Cluster-SMOTE + K-means, was proposed in [22], enhancing the robustness of benchmark evaluations. Simultaneously, the proliferation of industrial IoT has necessitated adaptive threat response mechanisms. As such, a novel threat-hunting framework utilizing Multiclass AdaBoost coupled with majority voting was delineated in [23],

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demonstrating superior performance over traditional models in detecting multiclass anomalies across diverse datasets. Moreover, the integration of PP measures within these systems has become paramount. The innovative PP anomaly detection cyber-physical system (PPAD-CPS) framework outlined in [24] employs transformative data reforming techniques to ensure privacy. Complementing this, [14] introduced "Block Hunter," an FL-based system tailored for Blockchain-based IoT networks, which impressively combines anomaly detection with reduced bandwidth usage by 35%, illustrating compatibility with edge computing devices [25]. This multifaceted approach not only fortifies data confidentiality but also underscores the evolving landscape of CPSS security solutions.

Furthermore, an amalgamation of FL with an Autoencoder, Transformer, and Fourier mixing sublayer gave rise to a robust distributed anomaly detection architecture. This architecture significantly boosted accuracy in CPSS [26]. In parallel, a cooperative intrusion detection model, which merged convolutional neural networks with gated recurrent units within an FL framework, was developed. This model was particularly noteworthy for its ability to maintain data privacy [27]. Expanding on this concept, an FL-based, self-regulating distributed system for intrusion detection was suggested, specifically targeting IoT devices [28]. In addition, an FL model was proposed to enable precise anomaly detection in time series data, showcasing its versatility [29]. In line with emerging trends in human-centric and multimodal FL, [30] proposed a model-contrastive learning approach to personalize FL in the Metaverse, optimizing multimodal user modeling. In healthcare IoT contexts, [31] introduced a hierarchical FL framework that incorporates social-context clustering for participant selection, enhancing efficiency and relevance in medical applications.

A significant advancement in model explainability for threat detection was achieved through the use of XAI, particularly SHAP, to analyze sensor data during attacks [32]. This approach was effective in rapidly identifying false alarms. Moreover, the employment of Decision Trees in explaining model results yielded an accuracy exceeding 95% [33]. The FedEx architecture, another notable innovation, exhibited exceptional performance in anomaly detection, quick learning of new patterns, and provided explainable results [34]. Finally, [35] presented a lightweight anomaly detection method using a reconstructed graph neural network enhanced by knowledge distillation, enabling efficient and scalable threat detection.

In addition to these developments, exploring security and privacy challenges in distributed computing continuum systems, as delineated in [36], offers a comprehensive perspective on the architecture and design principles of distributed systems. To mitigate data leakage risks, auditable PPFL (AP2FL) [37] was introduced. This method encompasses secure training and aggregation processes at both client and server levels using trusted execution environments (TEEs). This approach not only enhances accuracy but also robustly prevents privacy breaches. Furthermore, the research in [38] discusses dynamic resource scheduling in deterministic FL within time-sensitive networks. The authors established a multiobjective stochastic optimization framework that optimizes the success rate of scheduling and learning accuracy while adhering to deterministic constraints such as delay, jitter, and packet loss. This significantly improves the performance of FL. The study in [39] addresses the challenges of privacy preservation and reliability in decentralized FL. It introduces a novel approach that accommodates batch joining and client leaving with minimal delay while maintaining high model accuracy. The simulation results from this study validate the effectiveness of this strategy in balancing model accuracy and data privacy. Finally, [40] explores a federated K-nearest neighbors (KNNs)-based method with discrete coordinate encryption, ensuring the confidentiality of precise location coordinates for each target node. While this approach initially increases localization error, deploying dynamic access points using Hausdorff distance, especially within convex hull regions, significantly improves localization accuracy. In addition, [41] proposes a framework to modify data streams for privacy protection by implementing various policies at the data source. This framework enables the creation and articulation of high-level privacy policies in a user-friendly format, with an edge-based system examining data streams from adjacent IoT devices, applying these privacy policies, and modifying the data streams if privacy breaches are detected.

III. Problem Statement

A. Threat Model

Ensuring privacy and security in FL is essential, given potential threats. This section highlights risks in anomaly detection, particularly the server's potential as an adversary and the danger of client data breaches. While the server generally follows the protocol, its curiosity could compromise user privacy. In addition, the threat of data leaks from third-party sharing is significant. Based on these risks, we propose the following privacy measures. *1) Protection of Local Data*: In a CPSS setup for neural network training, clients share data with external entities. This sharing can inadvertently expose sensitive aspects of the client's operations. Training data must remain within the client's secure domain to counteract this. *2) Securing Gradients*: Given the potential for adversaries, such as a curious server, to exploit shared gradients, encrypting each user's local gradient before server transmission is vital. This step protects individual data and bolsters the overall CPSS security.

B. Problem Formulation

We address the challenge of crafting a privacy-centric, interpretable anomaly detection system for ICPS using FL. The goal is to detect anomalies in decentralized sensor data, ensuring data privacy and offering prediction explanations. Given N clients with sensors capturing multivariate time-series data, data from client n is represented as $D_n = [\mathbf{d}_n^1, \ldots, \mathbf{d}_n^{T_n}]$, where $\mathbf{d}_n^t$ is an M-dimensional vector of readings from M sensors at time t for client n. The model's main task is to label time instants in test data as normal or anomalous, represented by $A = [a_1, \ldots, a_{T_{\text{test}}}]$, with $a_t \in \{0, 1\}$. An anomaly at time t is denoted by $a_t = 1$, while $a_t = 0$ indicates normalcy. We use the

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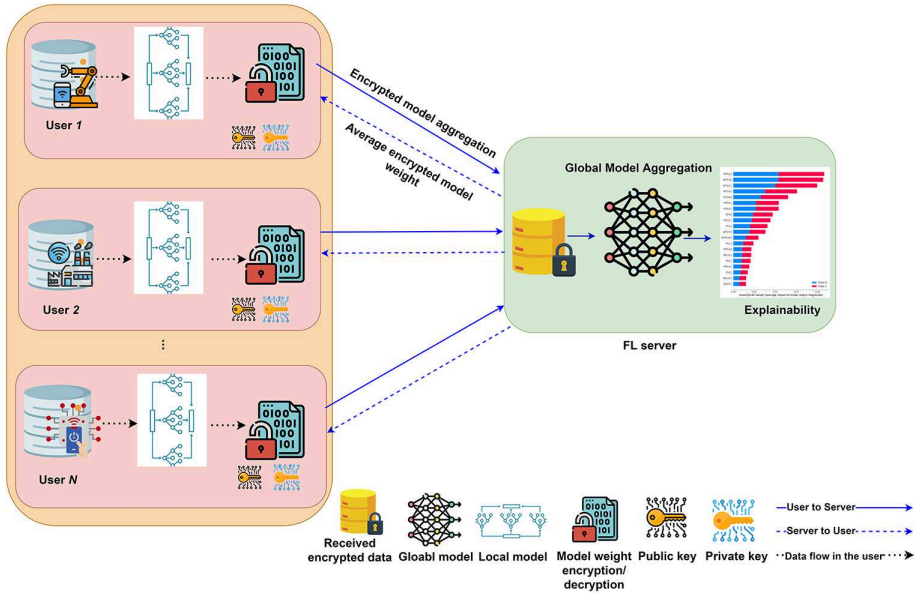


Fig. 1. Explainable PPFL framework in CPSS.

mean squared error (MSE) to measure the difference between model predictions and actual data $\mathbf{d}(t)$

$$\text{MSE} = \frac{1}{M}\sum_{t=1}^{M}(\hat{d}(t) - d(t))^2. \tag{1}$$

Training involves collaborative training of a global model G with parameters θ via FL. In our framework, we define the global model as the central model maintained and updated by the federated server through the aggregation of client updates. Each participating client maintains a local model, which is a replica of the global model initially received from the server, subsequently trained using the client's private data. During each communication round, the clients update their local models and send PP updates back to the server, which are then aggregated to refine the global model.

As training data only contains normal patterns, it is an unsupervised anomaly detection problem. The global model, $\theta_G = \sum_{i=1}^{N}(\theta_i)$, is derived from aggregated gradients in FL. Posttraining, the aim is to spot anomalies in test data, $D_{\text{test}} = [\mathbf{d}\text{test}^1, \ldots, \mathbf{d}\text{test}^{T\text{test}}]$, collected over a unique period. Each $\mathbf{d}_t^{\text{test}}$, akin to training data, is an M-dimensional vector of sensor readings at time t. This problem formulation seeks a balance between data privacy, interpretability, and performance in ICPS anomaly detection. While this formulation assumes a static training dataset, practical CPSS deployments often encounter temporal drift in data distributions. Addressing this challenge requires future extensions that incorporate dynamic model updates and concept drift handling mechanisms.

IV. Proposed Framework

We introduce a cutting-edge framework for an explainable PPFL model specifically crafted for threat detection in CPSS. This framework, illustrated in Fig. 1, is engineered to bolster reliability and security in CPSS operations against diverse cyber threats. It leverages the collective expertise of various participants from industry, cyber networks, and social domains, who iteratively refine their local models—copies of the global model—using their own datasets, and send encrypted updates to improve the global model maintained by the server, conducting training on their distinct datasets. Our framework employs PHE and DP to safeguard the transmission of model updates to the central server. This dual strategy ensures that gradient uploads remain confidential and integral, using a combination of public and private keys for encrypting and decrypting model parameters. By converting gradients into ciphertexts with a public key before aggregation and decrypting them at the server with a private key, the framework prevents unauthorized data access and exposure of sensitive model details, crucially reducing the risk of cyber intrusions.

In addition, to address the opacity often associated with ML models, we integrate SHAP into our system. This integration enhances transparency by enabling stakeholders to understand the influence of input features on model predictions, thereby increasing the trustworthiness and dependability of the threat detection process. In the context of CPSS, this transparency is essential, as it spans the physical sensors, cyber components, and social interactions, providing a holistic view of system behavior. Future subsections will delve deeper into the various components of our framework, highlighting their significance and utility in fortifying CPSS against emerging cyber threats and detailing how they collectively enhance the overall system reliability.

A. Handling Privacy Preservation

To enhance the privacy safeguards in our model, we have integrated advanced cryptographic methods, namely PHE and DP. Although FL naturally provides some privacy by keeping training data local, our findings deemed this inadequate for

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the strict privacy demands of CPSS [42]. As detailed in the threat model (Section III), we aimed for a robust method to protect local data and secure gradient updates in FL. PHE enhances this by allowing computations on encrypted data, which complements FL's requirement to maintain data privacy during aggregation. In addition, we use DP to add statistical noise to data, obscuring individual entries and boosting output privacy. This combination of PHE and DP is vital for a strong, comprehensive privacy framework that meets CPSS's rigorous standards.

In this article, we utilize the PHE scheme, which specifically allows for addition operations to be carried out on ciphertexts. The Paillier cryptosystem has two primary functions: key generation and encryption/decryption.

1) Key Generation: A pair of keys, a public key (n, g) and a private key (λ, μ), are generated where $n = pq$ for two large prime numbers p and q, g is an element of the set of integers modulo n^2 denoted $Z^*_{n^2}$, $\lambda = \text{lcm}(p-1, q-1)$, and $\mu = (L(g^\lambda \mod n^2))^{-1} \mod n$ with $L(x) = (x-1)/n$.

2) Encryption: Given a message m and public key (n, g), the ciphertext c is $c = g^m r^n \mod n^2$ for random r chosen from Z^*_n.

3) Decryption: Given ciphertext c and private key (λ, μ), the plaintext m is $m = L(c^\lambda \mod n^2) \cdot \mu \mod n$.

The additive homomorphic property of the Paillier cryptosystem ensures that the product of the ciphertexts (modulo n^2) decrypts to the sum of their corresponding plaintexts (modulo n). Formally, for plaintexts m_1, m_2 and their corresponding ciphertexts c_1, c_2, we have $D(c_1 \cdot c_2 \mod n^2) = (m_1 + m_2) \mod n$. This feature allows the secure aggregating of encrypted gradients in our proposed FL model, ensuring privacy during model update aggregation. To further safeguard our framework's privacy, we employ DP, which adds controlled statistical noise to data, balancing privacy and utility. During local training at each site, DP helps obscure exact data in potential breaches by modifying model parameters. By fine-tuning the noise level or privacy budget, we balance privacy and accuracy. A function f provides ϵ-DP if for all datasets D_1 and D_2 differing on at most one element, and all $S \subseteq \text{Range}(f)$

$$Pr[f(D_1) \in S] \leq e^\epsilon Pr[f(D_2) \in S]. \quad (2)$$

Here, ϵ is a privacy parameter that controls the amount of noise added. Smaller values of ϵ lead to more noise and, thus, greater privacy. In the context of our proposed model, DP is implemented during the local training phase of each participant's model. The output of the training algorithm (the model's parameters) is perturbed by adding Laplacian noise proportional to ϵ before being sent to the central server. This mathematical assurance of privacy prevents potential adversaries from exploiting the shared model parameters to infer information about the underlying training data, further strengthening the PP aspects of the proposed FL framework.

B. Handling Multimodal Data Heterogeneity

In CPSS environments, cyber, physical, and social data streams differ significantly in scale (e.g., bitrates versus temperatures), format (categorical logs versus continuous sensor signals), and dimensionality (e.g., metadata versus time series). To enable joint processing, we apply modality-specific preprocessing pipelines: numerical signals are z-score normalized, while categorical features (e.g., protocols and attack types) are one-hot or label encoded. Dimensionality reduction (via PCA or feature selection) is optionally applied to high-dimensional inputs. All preprocessed features are then concatenated into a unified input vector per sample. Local FL clients use this aligned representation to train homogeneous models across sites, enabling aggregation without structural conflicts. Future extensions may explore embedding-based architectures for late fusion of multimodal streams.

C. Addressing Explainability

Explainability is a crucial feature in any ML model, especially in sensitive sectors such as ICPS, where understanding the decision-making process can significantly aid in addressing threats efficiently. The traditional closed-box nature of ML models, however, can often impede this transparency. To tackle this opacity, we integrate SHAP into our proposed framework. SHAP is a game-theoretic approach that explains the output of any ML model. It connects optimal credit allocation with local explanations by using the classical Shapley values from cooperative game theory and their related extensions. Formally, given a game with N players, a coalition S of players, a payout function $v(S)$ that assigns a real-valued payout to each coalition S, a player i's Shapley value $\phi_i(v)$ is given by the following equation:

$$\phi_i(v) = \sum_{S \subseteq N \setminus i} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [v(S \cup i) - v(S)]. \quad (3)$$

This formula ensures that each player's contribution is calculated as an average marginal contribution over all possible coalitions. The Shapley values have some desirable properties.

1) *Efficiency*: The sum of the Shapley values of all players equals the total payout.
2) *Symmetry*: If two players contribute equally to all coalitions, their Shapley values are the same.
3) *Dummy Player:* If a player does not contribute to any coalition, their Shapley value is zero.
4) *Additivity*: If we have two games with payout functions v_1 and v_2, the Shapley value of a player in the game with payout function $v_1 + v_2$ is equal to the sum of the Shapley values of the player in the games with payout functions v_1 and v_2.

In our model, input features are such as "players," with the model's prediction serving as the "payout." We use SHAP values to measure the influence of each feature on a prediction. SHAP values quantify how significantly each feature affects the outcome; those with higher absolute values are more impactful. This approach not only clarifies how decisions are made, enhancing transparency and trust in the model's predictions, but also identifies key features for threat detection. In the context of CPSS, this transparency spans the cyber, physical, and social domains, ensuring that critical features from all aspects

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Algorithm 1: Integrated Framework Procedure.

```
procedure IntegratedFramework
Require: Global model, local datasets, public and private keys (if PHE is used)
Ensure: Updated global model, SHAP values
    for each participant do
        Initialize local model as a copy of global model
        Train local model on participant's data
        if DP is chosen then
            Add noise to the gradients
        else if PHE is chosen then
            Encrypt gradients using public key
        end if
        Send modified gradients to server
    end for
    if PHE is chosen then
        Decrypt received gradients using a private key
    end if
    Aggregate received gradients
    Generate updated global model
    for each prediction do
        Compute SHAP values for each feature
    end for
    return Updated global model, SHAP values
end procedure
```

of the system are considered for threat detection. This informs subsequent data collection and model improvements, making our framework more effective and trustworthy for practical CPSS applications.

We outline the workflow of our framework, emphasizing its integration of FL, DP or PHE, and SHAP for PP explainable CPSS threat detection. The procedure unfolds in Algorithm 1. This procedure iteratively updates the global model using local data, safeguarding sensitive information from the central server. It also makes decision-making transparent, highlighting key features from cyber-physical-social interactions for effective threat response.

While SHAP was selected for its solid theoretical grounding and compatibility with tree-based models, alternative methods such as LIME and integrated gradients were also considered. However, the LIME perturbation-based approach can yield inconsistent results, and integrated gradients require unsuitable gradient access for our ensemble model, making Tree SHAP the most appropriate choice to balance interpretability, stability, and computational efficiency in CPSS settings.

V. Security Analysis

A. Security Analysis via PHE

The security of our protocol, which uses PHE, can be formally shown by defining a game between a challenger and an adversary. Let $\mathcal{A}$ be a polynomial time (PPT) adversary that can adaptively send queries to a decryption oracle $\mathcal{O}_{\text{Dec}}$.

Definition (Game G1: Privacy Game): Let A be an adversary that has access to a privacy mechanism M. The game is defined as follows: 1) The adversary A selects datasets (D_0 and D_1). 2) A random bit $b \in \{0, 1\}$ is chosen. The mechanism M is applied on D_b and the output O is given to the adversary. 3) The adversary A makes a guess b'. If $b = b'$, the adversary wins the game.

Now, we can say that the privacy mechanism M is (ϵ, δ)-differentially private if for all adversaries A and all possible outputs O, the following holds:

$$|\Pr[A(M(D_b)) = 1] - \Pr[A(M(D_{1-b})) = 1]| \leq e^{\epsilon} + \delta \quad (4)$$

where probabilities are over the random coins of A and M.

Game G1: The challenger generates a pair of keys (pk, sk) using the PHE key generation algorithm $Gen(1^k)$, where k is the security parameter. 1) The challenger sends pk to $\mathcal{A}$. 2) $\mathcal{A}$ sends two equal-length messages m_0, m_1 to the challenger. 3) The challenger selects a random bit $b \leftarrow 0, 1$, encrypts m_b using pk to get $c = Enc(pk, m_b)$, and sends c to $\mathcal{A}$. 4) $\mathcal{A}$ can make polynomially many queries to $\mathcal{O}_{\text{Dec}}$ (excluding the challenge ciphertext c), and eventually outputs a bit b'. The advantage of $\mathcal{A}$ in the game S is defined as $|Pr[b' = b] - 1/2|$. The PHE scheme is semantically secure if, for any PPT adversary $\mathcal{A}$, the advantage of $\mathcal{A}$ in the game G1 is negligible.

B. Security Analysis via DP

We analyze the security of our protocol under the framework of DP. Given a dataset, D, a function f, and a privacy parameter ϵ, a randomized algorithm M satisfies ϵ-DP if, for any D and D' that differ by at most one entry and for any possible output O of $M(D)$, we have $(\Pr[M(D') = O]/\Pr[M(D) = O]) \leq e^{\epsilon}$. This condition ensures that the inclusion or removal of any individual data entry does not significantly change the probability distribution of the output, thus providing privacy guarantees. To ensure the ϵ-DP, our model adds Laplace noise, which is drawn from the Laplace distribution $\text{Lap}((\Delta f/\epsilon))$, where Δf is the sensitivity of function f, to the output of function f over the dataset. The sensitivity of a function f, Δf, is the maximum difference in the function outputs on any two datasets that differ by exactly one entry, $\Delta f = \max_{D,D':|D-D'|=1} |f(D) - f(D')|$. Our model, by using PHE and DP together, ensures that the privacy of individual data is protected against any PPT adversaries and the release of statistical information is controlled under a predefined privacy budget. This provides a robust PP mechanism for using personal data in computing environments.

C. Adversarial Risks to Explainability

While our model enhances transparency using SHAP, we acknowledge that adversarial attacks can target the explanation layer itself. For example, model inversion or feature importance manipulation may lead to misleading or obfuscated explanations. Although our focus in this study is on securing the training and aggregation processes via DP and PHE, addressing attacks on interpretability modules is an open challenge. Future work will explore robust explainability techniques that can resist adversarial influence and ensure trustworthy insight generation.

VI. Performance Evaluation

Our experiments were run on a high-performance computing server equipped with an Intel(R) Core(TM) i9-7900X CPU clocking at 3.30GHz (Xeon), supplemented by a powerful

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GeForce RTX 2080 Ti GPU. The system was further enhanced with 128 GB of RAM to handle data-intensive computations. The software environment constituted Python 3.10, and we leveraged the functionalities of the PyTorch/torch libraries to implement the proposed framework. We employed the python-*Paillier library* [43] with a 1024-bit public key for secure data encryption. DP was implemented using Diffprivlib v0.6. The phe library was used for performing computations on encrypted data through homomorphic encryption. Last, the SHAP library was utilized to provide model interpretability. Collectively, these libraries enabled us to establish a secure, private, and explainable framework capable of effectively handling sensitive data. Our evaluation of the proposed model employed three distinct datasets: SWaT[1], Gas Pipeline[2], and UNSW-NB15[3], each offering unique data structures and characteristics that are relevant for CPSS.

The SWaT dataset, compiled by the Singapore University of Technology and Design, contains 449 919 samples across 52 features [44]. This dataset captures critical sensor data from a water treatment system, making it highly representative of the physical layer in CPSS. For evaluation purposes, we divided the dataset into a training set (70%) and a testing set (30%).

The Gas Pipeline dataset provides data on a gas pipeline system, comprising 97 019 instances across 26 features [45]. This dataset reflects both the physical infrastructure and cyber operations within the pipeline system, contributing valuable insights into the interaction between cyber and physical components. Similarly, we utilized 70% of the data for model training and reserved 30% for testing.

Last, the UNSW-NB15 dataset, curated by the Cyber Range Lab of the Australian Centre for Cyber Security (ACCS) at UNSW, offers a comprehensive dataset with a total of 257 673 rows and 42 features [46]. This dataset primarily represents network traffic data, which is crucial for understanding the cyber-social interactions in CPSS, particularly in terms of detecting network-based attacks and anomalies. By utilizing these datasets, we can ascertain the robustness and effectiveness of our proposed model across varied data structures and domains within the broader context of CPSS, encompassing cyber, physical, and social dimensions.

1) Evaluation Metrics and Model: To assess our model's performance, we employed key metrics: F1 score, recall, and precision. These metrics gauge the model's accuracy in distinguishing between positive and negative instances. The metrics are defined as: Accuracy(Acc) $=$ (TP $+$ TN/TP $+$ TN $+$ FP $+$ FN), Precision(Pre) $=$ (TP/TP $+$ FP), Recall(Rec) $=$ (TP/TP $+$ FN), and $F1 = 2 \times (Pre \times Rec/Pre\#x002B; Rec)$. Here, TP and TN denote correct identifications of positive and negative instances. FP and FN represent misclassifications. These metrics offer a holistic view of our model's performance. In our FL framework, we used a RandomForestClassifier with parameters: 20 trees and a maximum depth of 10. We engaged $10 - 10^3$ clients over 10 FL rounds, aiming for optimal performance within these constraints.

[1]https://itrust.sutd.edu.sg/itrust-labs_datasets/dataset_info/
[2]https://www.impactcybertrust.org/dataset_view?idDataset=1322
[3]https://research.unsw.edu.au/projects/unsw-nb15-dataset

TABLE I
RESULT OF SWAT DATASET

Method	Acc (%)	Pre (%)	Rec (%)	F1 (%)
FL	0.9746	0.9978	0.7927	0.8835
FL with PHE	0.9746	0.9978	0.7927	0.8835
FL with DP	0.6686	0.2065	0.6097	0.3086
FL with PHE + DP	0.7812	0.4157	0.6510	0.5065
Ensemble SAE & DNN [47]	0.9967	0.97	0.99	0.99
SVM [48]	-	0.93	0.699	0.79
TABOR [49]	0.9499	0.86	0.79	0.82

TABLE II
RESULT OF GAS PIPELINE DATASET

Method	Acc (%)	Pre (%)	Rec (%)	F1 (%)
FL	0.9885	0.9918	0.9768	0.9842
FL with PHE	0.9885	0.9918	0.9768	0.9842
FL with DP	0.66706	0.5375	0.6588	0.5920
FL with PHE + DP	0.8413	0.6768	0.7451	0.7093
Ensemble SAE & DNN [47]	0.96	0.9463	0.9372	0.9383
SVM [50]	0.92	0.7820	0.9360	0.8520
DT [47]	0.86	0.9159	0.6808	0.7239

A. Rationale for Dataset Selection

The three selected datasets represent a comprehensive cross-section of CPSS domains. The SWaT dataset reflects physical process control with sensor-driven data from industrial water treatment systems, ideal for modeling physical-layer threats. The Gas Pipeline dataset bridges both physical infrastructure and cyber monitoring, providing insights into pipeline telemetry and control data. The UNSW-NB15 dataset, although widely used for network security, also captures behavioral traffic patterns such as access frequencies, usage profiles, and attack interactions, reflecting the human and social dimension of CPSS. Its inclusion allows us to evaluate the model's ability to detect anomalies rooted in user behaviors and network-layer social interactions, thus completing the triad of cyber, physical, and social components central to CPSS.

B. Results and Analysis of the Proposed Model in CPSS

This part of our work presents the performance metrics obtained from our anomaly detection experiments. We perform anomaly detection under three different PP conditions: *FL*, *FL with PHE*, and *FL with DP*. We benchmark our results against other research work using three distinct datasets: *SWaT*, *Gas Pipeline*, and *UNSW-NB15*. The outcomes are presented in Tables I–III.

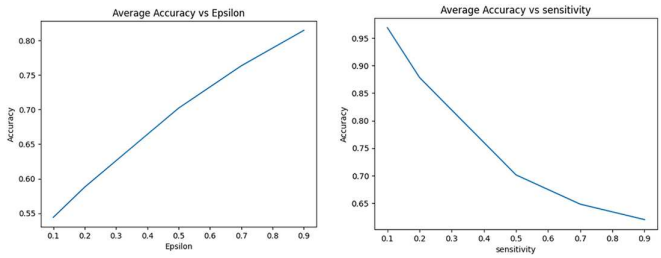
For all three datasets, we found that FL and FL with PHE yielded identical performance metrics, proving that PHE preserves privacy without compromising anomaly detection accuracy, though at a higher computational cost. Conversely, FL with DP reduces computational overhead compared to PHE but at the expense of performance metrics. Notably, our PP methods outperformed non-PP models in the Gas Pipeline dataset, highlighting the balance between privacy and performance. With FL and DP, we set the sensitivity to 0.5 and epsilon to

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TABLE III
RESULT OF UNSW-NB15 DATASET

Method	Acc (%)	Pre (%)	Rec (%)	F1 (%)
FL	0.9360	0.9502	0.9495	0.9499
FL with PHE	0.9360	0.9502	0.9495	0.9499
FL with DP	0.6493	0.7614	0.6574	0.7056
FL with PHE + DP	0.7935	0.8347	0.6992	0.7603
NAFT [51]	0.9723	0.9748	0.9696	-
DNN [51]	0.9374	0.9437	0.9302	-
LR [51]	0.8527	0.8513	0.8546	-
K-NN [51]	0.8372	0.8316	0.8456	-



(a) (b)

Fig. 2. Average accuracy versus epsilon and sensitivity in DP. (a) Epsilon versus accuracy. (b) Epsilon versus sensitivity.

0.4, resulting in lower performance but enhanced privacy. Our study also explored how DP parameters affect accuracy. Generally, increasing epsilon reduces privacy, while higher sensitivity boosts it. Fig. 2, derived from all datasets, visually represents these dynamics, showing how accuracy varies with changes in epsilon (at constant sensitivity) and how accuracy shifts with alterations in sensitivity (at constant epsilon).

We further evaluated the joint application of both PHE and DP in our FL setup (denoted as "FL with PHE + DP"). The results, added to Tables I–III, indicate a noticeable decline in performance compared to using either privacy technique alone. Specifically, the addition of DP's statistical noise atop PHE's encrypted gradient aggregation introduces compounded perturbations that affect detection accuracy and precision. However, the results still demonstrate acceptable levels of F1-score and recall, especially in the UNSW-NB15 dataset. This tradeoff emphasizes the critical balance between stringent privacy guarantees and detection efficacy in high-risk CPSS environments.

To further analyze the privacy-utility tradeoff, we plotted precision across multiple ϵ values (ranging from 0.1 to 1.0) for each dataset, shown in Fig. 3. The results confirm a consistent pattern: increasing ϵ—thereby reducing the strength of DP—results in higher precision. For instance, in the SWaT dataset, precision rises from 0.12 ($\epsilon = 0.1$) to 0.46 ($\epsilon = 1.0$). Similar trends are observed for Gas Pipeline and UNSW-NB15, with final precision values reaching 0.76 and 0.90, respectively. This visual analysis emphasizes the sensitivity of the detection performance to the DP budget and helps to select ϵ values that balance privacy constraints with detection quality.

1) Impact of DP Noise on Precision: We conducted additional experiments to assess how different DP noise levels (ϵ values) affect anomaly detection precision. As expected,

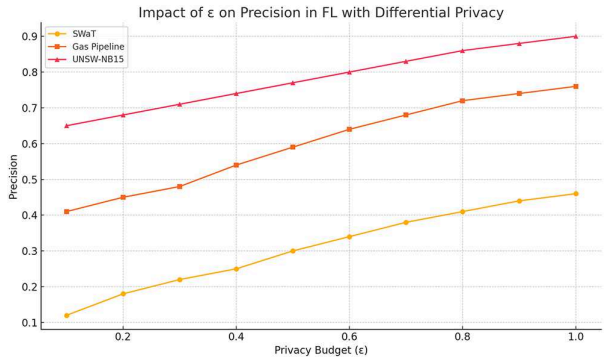


Fig. 3. Precision as a function of the privacy budget ϵ in FL with DP across three CPSS datasets. Higher ϵ values correspond to weaker privacy but improved model performance.

stronger privacy (lower ϵ) introduces more noise, reducing precision. For example, on the SWaT dataset, precision dropped from 0.4093 ($\epsilon = 0.8$) to 0.2065 ($\epsilon = 0.4$), while on Gas Pipeline, it fell from 0.6721 to 0.5375 across the same ϵ range. This confirms that DP's statistical noise affects the model's ability to correctly identify true positives, especially under tighter privacy constraints.

2) Tuning FL Parameters for New Datasets: To adapt our framework to new CPSS datasets, we follow a practical tuning process. First, we perform an offline grid search on a small validation set to determine a suitable learning rate, typically exploring values between 0.001 and 0.05. We then vary the number of communication rounds (10–100) and assess convergence behavior and performance metrics (accuracy, precision, and F1). The number of local training epochs is also adjusted (1–5) to balance computation and communication costs. Finally, we evaluate performance across varying numbers of clients to account for data heterogeneity. In our experiments, we observed that while larger datasets slightly increased convergence time, the model maintained stability and achieved consistent performance across all three datasets, demonstrating scalability in training dynamics.

C. Complexity and Computational Analysis

The proposed FL model incorporates DP and PHE to enhance privacy without significantly compromising the system's real-time responsiveness [52]. This subsection delineates the computational analysis of the proposed model, highlighting the impact of these PP methods on the computation overhead. Within the realm of FL, the complexity of implementing DP and PHE mechanisms is multifaceted, potentially impacting real-time performance due to computational overhead, especially in the proposed model. This subsection outlines the computational and communicational burdens these mechanisms impose and how they are managed within our proposed model that encompasses a server and multiple clients.

1) *Client-Side Complexity*: On the client side, the incorporation of DP involves the addition of controlled noise to data or gradients before sharing them with the server, a process that incurs a computational complexity of $\mathcal{O}(d)$ for data

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TABLE IV
COMPUTATION SUMMARY OF THE PROPOSED MODEL WITH DP AND PHE

FL Components	Computation Without DP and PHE	Computation With DP and PHE	Overhead
User	$\mathcal{O}(d \cdot N)$	$\mathcal{O}((d+p) \cdot N \cdot \log N)$	$\mathcal{O}(p \cdot N \cdot \log N)$
Server	$\mathcal{O}(d \cdot N \cdot \log N)$	$\mathcal{O}((d+p) \cdot N \cdot \log N)$	$\mathcal{O}(p \cdot N \cdot \log N)$
Overall System	$\mathcal{O}(d \cdot N \cdot \log N)$	$\mathcal{O}((d+p) \cdot N \cdot \log N + N \cdot d \cdot \log d)$	$\mathcal{O}(p \cdot N \cdot \log N + N \cdot d \cdot \log d)$

with dimension d. Using PHE, specifically the Paillier cryptosystem, adds another layer of computational cost due to encryption operations. The complexity of encryption for each client is $\mathcal{O}(d \cdot \log d)$, performed once per training iteration for each client.

2) *Server-Side Complexity*: At the server, the complexity of aggregating encrypted updates from clients using PHE is $\mathcal{O}(N \cdot d \cdot \log N)$, where N is the number of clients and d is the data dimension. The decryption of the aggregated update incurs a complexity of $\mathcal{O}(d \cdot \log d)$. For DP, the server must also calculate the appropriate scale of the noise to be added to the aggregated model update, a process that adds a minor computational overhead proportional to the number of model parameters [52].

3) *Communication Overhead*: Communication overhead arises from the need to transmit encrypted updates from clients to the server and the distribution of the updated model back to clients. The size of the encrypted data is proportional to the key size used in PHE, and hence, the communication cost increases linearly with both the number of clients N and the data dimension d. The detailed computational costs associated with each component of the FL model, considering the implementation of DP and PHE, are summarized in Table IV. This table illustrates the overhead introduced by incorporating these PP methods into our system. It highlights the balance we have struck between enhancing privacy and maintaining the system's real-time responsiveness.

1) Communication Overhead of PHE: In addition to computation latency, PHE introduces a nonnegligible communication overhead. Specifically, in the Paillier cryptosystem, each encrypted value is represented as a large integer of size roughly equal to the modulus (e.g., 1024 and 2048 bits). In our setup, encrypting 32-bit floating-point gradients results in ciphertexts approximately 64× larger, increasing the payload size for each model update. For instance, transmitting a 10 KB plaintext model update would result in a 640 KB encrypted payload under 2048-bit keys. Although such payload sizes are feasible in wired or high-bandwidth industrial networks, they may pose challenges in bandwidth-constrained environments. We suggest compressing model updates or reducing gradient precision as practical mitigations for communication-limited deployments.

D. Computational Time and Resources

In the proposed model, computational time and resources are crucial metrics for evaluating the feasibility of PP techniques such as DP and PHE. For PHE, specifically the Paillier cryptosystem, the computational overhead is directly tied to the key sizes chosen for encryption and decryption. Table V showcases the overheads for various key sizes, revealing a trend where larger keys, though providing enhanced security, correspond to increased computational time for operations. In the context of DP, the added noise to ensure privacy incurs additional computational costs, primarily during the noise generation and aggregation processes. While not as computationally intensive as PHE, the complexity of DP can be characterized as $\mathcal{O}(n \cdot m)$, where n is the number of data points and m is the number of parameters to which noise is added. The computational overhead for DP is also influenced by the privacy budget, with tighter privacy constraints (lower ϵ-values), often leading to higher computational costs due to more complex noise calculations.

TABLE V
OPERATION TIMES BASED ON VARIOUS KEY SIZES

Key Size (bit)	Operation Time (# Operations Per Second)		
	Encrypt	Decrypt	Generating Keypair
128	0.000134 s (8551 ops/s)	0.00052 s (11 009 ops/s)	0.001 S
256	0.000309 s (2508 ops/s)	0.000171 s (5007 ops/s)	0.02 S
512	0.002250 s (300 ops/s)	0.000582 s (1007 ops/s)	0.01 S
1024	0.011911 s (61 ops/s)	0.003607 s (117 ops/s)	0.1 S
2048	0.088179 s (8 ops/s)	0.020408 s (25 ops/s)	0.4 S

Note: All runtimes are reported in milliseconds. These values represent average per-operation latencies for encryption, decryption, and keypair generation using the Paillier cryptosystem on a 3.3 GHz CPU.

1) Computational Tradeoffs of Explainability: While SHAP provides model-agnostic and theoretically grounded interpretability, it comes with computational costs, particularly for high-order models or large input feature spaces. Kernel SHAP, for instance, has exponential complexity in the number of input features, making it impractical for real-time analysis in large-scale deployments. To mitigate this, we utilized Tree SHAP for our Random Forest models, which scales polynomially and provides exact solutions efficiently. Moreover, SHAP explanations were computed posthoc on representative samples rather than on all test instances to limit overhead. This strategy ensures a balance between transparency and runtime performance, which is essential for CPSS environments where both accuracy and responsiveness are critical.

To contextualize the practical cost of PHE, Table V shows per-operation latencies across key sizes. For the 1024-bit key used in our experiments (chosen as a standard tradeoff between security and performance), encryption and decryption took approximately 11.9 and 3.6 ms per vector, respectively. Keypair generation was completed in 100 ms. Given that encryption is performed once per client per communication round, and decryption once per server aggregation, these latencies remain tractable in FL setups with periodic updates (e.g., every few seconds and minutes in CPSS). Even for 2048-bit keys, while encryption time increases to 88 ms, the overall impact on detection latency is moderate, making PHE viable for near-real-time anomaly detection in industrial systems.

2) Impact of Key Length on Security and Latency: To quantify the tradeoffs between security strength and system

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TABLE VI
MEMORY USAGE FOR DP OPERATIONS

Privacy Budget (ϵ)	Data Size	Memory Usage (MB)	Noise Generation Time (s)
1.0	1K records	1.2	0.05
0.5	1K records	2.4	0.10
1.0	10K records	12	0.5
0.5	10K records	24	1.0
1.0	100K records	120	5
0.5	100K records	240	10

performance, we evaluated Paillier encryption with key sizes of 512, 1024, and 2048 bits (Table V). Each increase in key length approximately doubles ciphertext size and significantly impacts latency. For example, encryption latency increases from 2.9 ms (512-bit) to 88.3 ms (2048-bit), while decryption time rises from 1.2 to 25.7 ms. From a security standpoint, 1024-bit keys correspond to 80-bit symmetric security, which is considered the minimum acceptable threshold for short-term protection, while 2048-bit keys offer 112-bit security, aligning with NIST recommendations for medium-term use. In time-sensitive FL deployments, 1024-bit keys offer a balanced compromise between computational efficiency and cryptographic strength.

For both DP and PHE, the selection of parameters—whether the key size in PHE or the privacy budget in DP—must be judiciously balanced against the desired level of security and the allowable computational budget.

Memory usage is a significant concern in implementing DP, especially when managing the overhead of storing noise-added data or gradients and auxiliary structures for DP mechanisms. This overhead grows with high-dimensional data or when maintaining multiple perturbed data versions to meet different privacy levels. For example, splitting the privacy budget across various queries necessitates separate noise generation and storage for each, leading to a potential linear increase in memory needs relative to the number of queries. In addition, the precision of noise (such as using floating-point versus integer representations) influences memory consumption. Although our model is efficient in memory management, integrating DP can raise memory demands, which is an important consideration in system design, especially in resource-limited settings. Table VI illustrates this: a smaller privacy budget (ϵ) demands more noise, increasing memory usage and noise generation time. The "Data Size" column shows the data scale, while "Memory Usage" indicates the memory needed for storing the noisy data. "Noise Generation Time" reflects the time required for noise computation and addition.

E. Scalability of the Proposed Model

Scalability is a key issue in FL systems, especially with increasing numbers of clients and data volume. Our model is crafted for scalability, effectively handling more clients and larger datasets.

1) *Client Participation*: It performs well in client participation, balancing accuracy and privacy without significantly raising computational costs. As client count N grows, the model uses PHE for aggregation, keeping server-side computation cost linear ($\mathcal{O}(N \cdot d \cdot \log N)$) instead of exponential, vital for scalability. DP also scales well, independent of client numbers, and maintains privacy no matter the N.

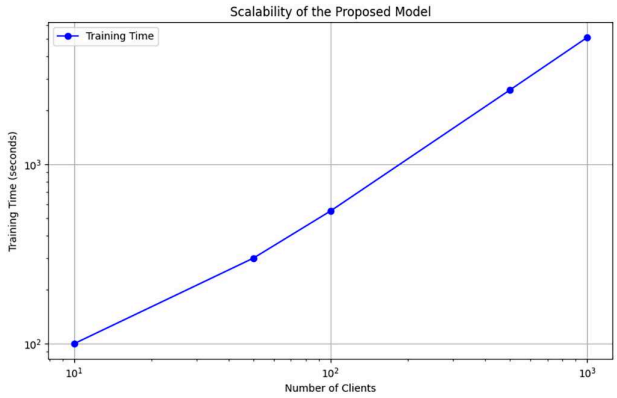


Fig. 4. Scalability of the proposed model with varying numbers of clients.

2) *Data Volume*: Regarding data volume, our model's federated structure allows it to thrive. Data is processed locally at each client, avoiding the need for increased central processing or storage with growing data volume. This distributed processing supports larger datasets by parallelizing computational load across clients.
3) *Communication Overhead*: To tackle communication overhead, crucial for scalability, the model uses efficient encoding to reduce update sizes transmitted between clients and the server. This approach is essential for managing network bandwidth issues with more clients. Employing model compression techniques such as sparsification and quantization further cuts down data transmitted, enhancing the model's scalability in communication.
4) *Model Evaluation and Testing*: Empirical evaluations have been conducted to test the scalability of the proposed model. Simulations with varying numbers of clients, ranging from a small-scale network of ten clients to a large-scale network involving thousands, have been executed. As depicted in Fig. 4, the model preserves its efficacy and efficiency, with only a marginal increase in training time and resource utilization. These findings are corroborated by tests on datasets of increasing sizes, from gigabytes to terabytes, demonstrating the model's ability to scale without significant degradation in performance.

F. Comparison With Related Works

Here, we discuss the comparative performance of our approach, as presented in Tables I–III, with other existing works. The results from the previous tables offer a comprehensive overview of our method's performance across different datasets. Across the three datasets—SWaT, Gas Pipeline, and UNSW-NB15—our FL and FL with PHE techniques consistently outperform or show comparable performance with other models regarding accuracy, precision, recall, and F1-score. For instance, with the SWaT dataset, our methods significantly outperformed SVM and matched closely with the Ensemble SAE and DNN model's results. A similar pattern is evident in the Gas Pipeline dataset, where we see a clear edge in performance over SVM and DT. Despite the slight performance decline with DP, it's

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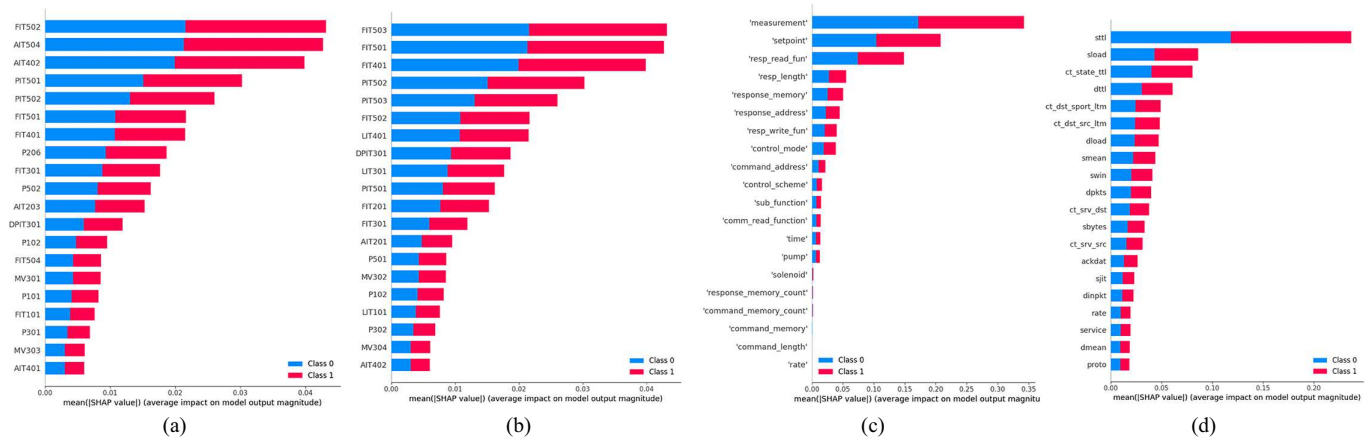


Fig. 5. SHAP-based feature importance analysis across three datasets: (a) results of explainability FL and FL + HE in SWaT; (b) results of explainability FL + DP in SWaT; (c) results of explainability FL, FL + HE, and FL + DP in Gas Pipeline; and (d) results of explainability FL, FL + HE, and FL + DP in UNSW-NB15. These summary plots depict the mean SHAP value of each feature across predictions, representing their contribution magnitude to model outputs. Red (Class 1) and blue (Class 0) indicate the distribution of feature impacts for anomalous and normal samples, respectively.

important to note that this tradeoff ensures enhanced privacy, a unique feature not offered by the other methods. Importantly, our work guarantees privacy without compromising model performance, distinguishing it from existing works. This approach holds significant value in real-world applications where both performance and privacy are crucial. In addition to performance and privacy, our work also emphasizes the explainability of our models.

G. Explanation in Anomaly Detection

Here, we elucidate the interpretability of our model by analyzing the significance of each feature, aiding us in understanding the model's decision-making process. Fig. 5 presents SHAP-based summary plots for the three CPSS datasets, visualizing the most influential features for threat classification under different privacy settings. These plots enable us to interpret the model's behavior by showing which input features most significantly influenced the classification of samples as either normal (Class 0, blue) or anomalous (Class 1, red). In the SWaT data set [Fig. 5(a) and (b)], key physical process indicators such as "FIT502," "AIT504," and "AIT402" exhibit high SHAP values, reinforcing their relevance in detecting anomalies of water treatment. In particular, when DP is applied [Fig. 5(b)], the importance distribution changes slightly, with "LIT401" and "DPIT301" gaining prominence, likely due to DP-induced perturbation.

In the Gas Pipeline dataset [Fig. 5(c)], characteristics such as *measurement, setpoint, and* $resp_read_fun$ dominate in all privacy settings, reflecting the basic telemetry and control flow variables. These high-impact fields align with expected points of cyber-physical interaction in the system. For the UNSW-NB15 dataset [Fig. 5(d)], SHAP identifies *sttl*, *sload*, and ct_state_ttl as the main contributors, characteristics associated with network behavior and packet timing at session level. These findings highlight that the model accurately emphasizes features aligned with known cybersocial attack vectors. Importantly, across all datasets, the SHAP profiles of FL and FL with PHE remain nearly identical, confirming that PHE preserves model interpretability. The FL with DP configuration exhibits slight variations due to DP noise but retains the overall feature importance hierarchy. This consistency supports the conclusion that, when properly configured, PP mechanisms do not substantially compromise interpretability in anomaly detection tasks.

VII. Conclusion

We presented an innovative and explainable PPFL model for CPSS, addressing the growing demand for secure, reliable, and interpretable anomaly detection. Our approach leverages FL while integrating DP and PHE to ensure strong privacy during training and update aggregation. Experiments on CPSS datasets confirm the model's superior performance over baselines, highlighting its effectiveness across cyber, physical, and social layers. It maintains a strong balance between privacy, interpretability, and detection accuracy, achieving high performance without sacrificing explainability. SHAP-based integration enhances model transparency and enables meaningful threat interpretation, building trust in automated CPSS responses. We also empirically assessed the joint integration of PHE and DP and observed that although it results in moderate degradation in detection precision, it provides stronger cumulative privacy guarantees, relevant in scenarios demanding both communication encryption and differential protection of model updates. While this study focuses on static privacy and aggregation settings, dynamically tuning these parameters via reinforcement learning (RL) could improve adaptability, though at the cost of added complexity, which we leave for future work. Future extensions could also incorporate continual or online FL to adapt to evolving CPSS environments and mitigate concept drift, ensuring robust real-time detection. Finally, as interpretability frameworks such as SHAP face emerging adversarial threats, exploring robust and certifiable explanation methods will be a critical next step.

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